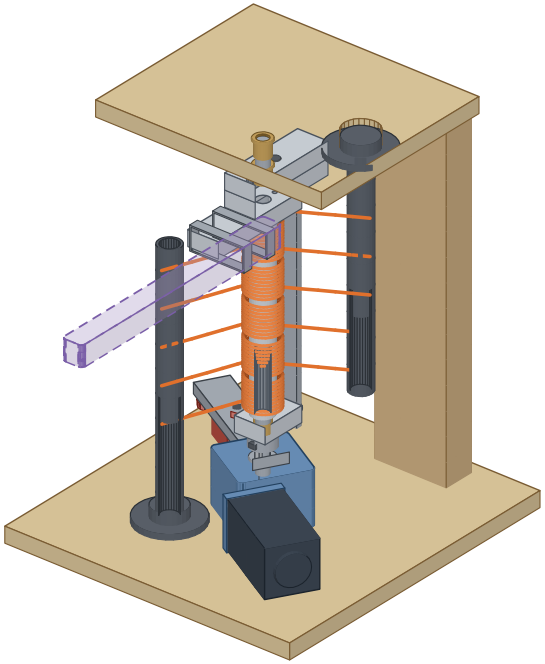
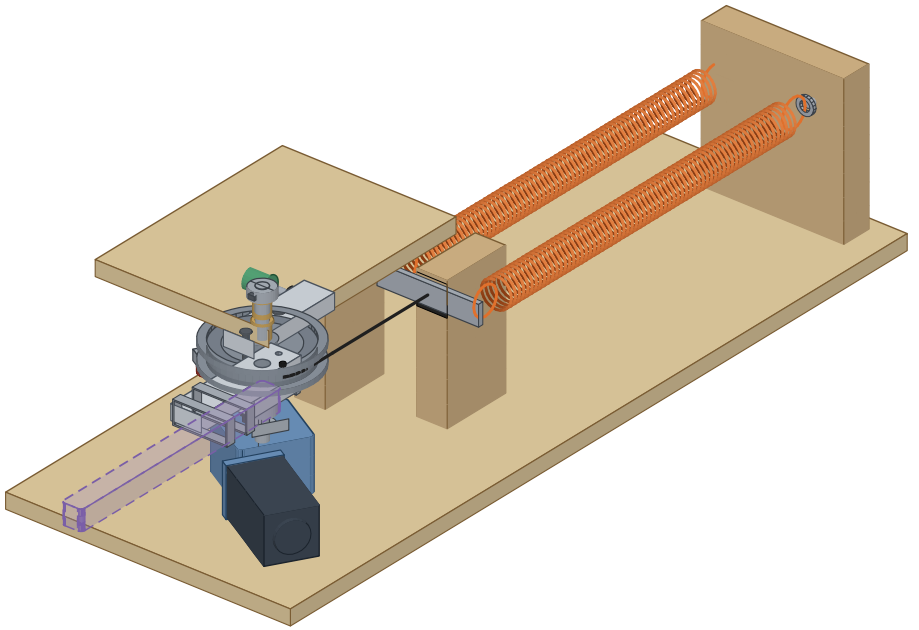
Two layouts, side by side at the same scale

Option 1 torsion pack · Option 2 extension springs
Both: NEMA 23 + NMRV030 10:1, 20 m/s at 120°



Option 1 · 44 cm tall



Option 2 · 24 cm tall · paddle dashed

	OPTION 1 · TORSION PACK	OPTION 2 · EXTENSION SPRINGS
Height	**44 cm** (17.3 in)	**24 cm** (9.3 in)
Base footprint	14 × 15 in (356 × 381 mm)	14 × 37 in (356 × 939 mm); the springs lie flat behind the shaft
Springs	5 McMaster torsion springs (9271K132) stacked on the shaft	2 Century Spring extension springs (660CS) pulling a cord on a 5 in V-pulley
Top setting (120°)	21.2 N·m spring, 2.49 N·m motor, 20 m/s tip	20.2 N·m spring, 2.38 N·m motor, 20 m/s tip
Est. cost, all new	≈ $300 (≈ $243 with plywood, lumber and hardware on hand)	≈ $286 (≈ $219 with plywood, lumber and hardware on hand)
Good	Small footprint. Springs are carried by the hub, no cord to wear	Low. Only 2 springs. Preload is set by stretching springs, no winding
Watch out	Taller. Each spring is wound about 145° during assembly (about 6 lb at the leg end; go carefully)	Long base. Dyneema knots need care. The spring stop and anchor take about 60 lb all the time

Same in both: the paddle axis is vertical and the paddle swings flat. Both use the same release (drive arm keyed to the gearbox, servo-tripped hook, free-spinning paddle hub). So do the gearbox, motor, shaft size and bushings.

Stop: a **stop block** sits on top of the hub, and the absorber hangs from the top plate **above the paddle**. The absorber's push goes straight into the top plate instead of bending the shaft or a tall post.

Both layouts were swept through every angle from the overrun to 120° with a worst-case 8 in wide blade. Nothing at paddle height sits inside the swing, and the springs never touch the hub.

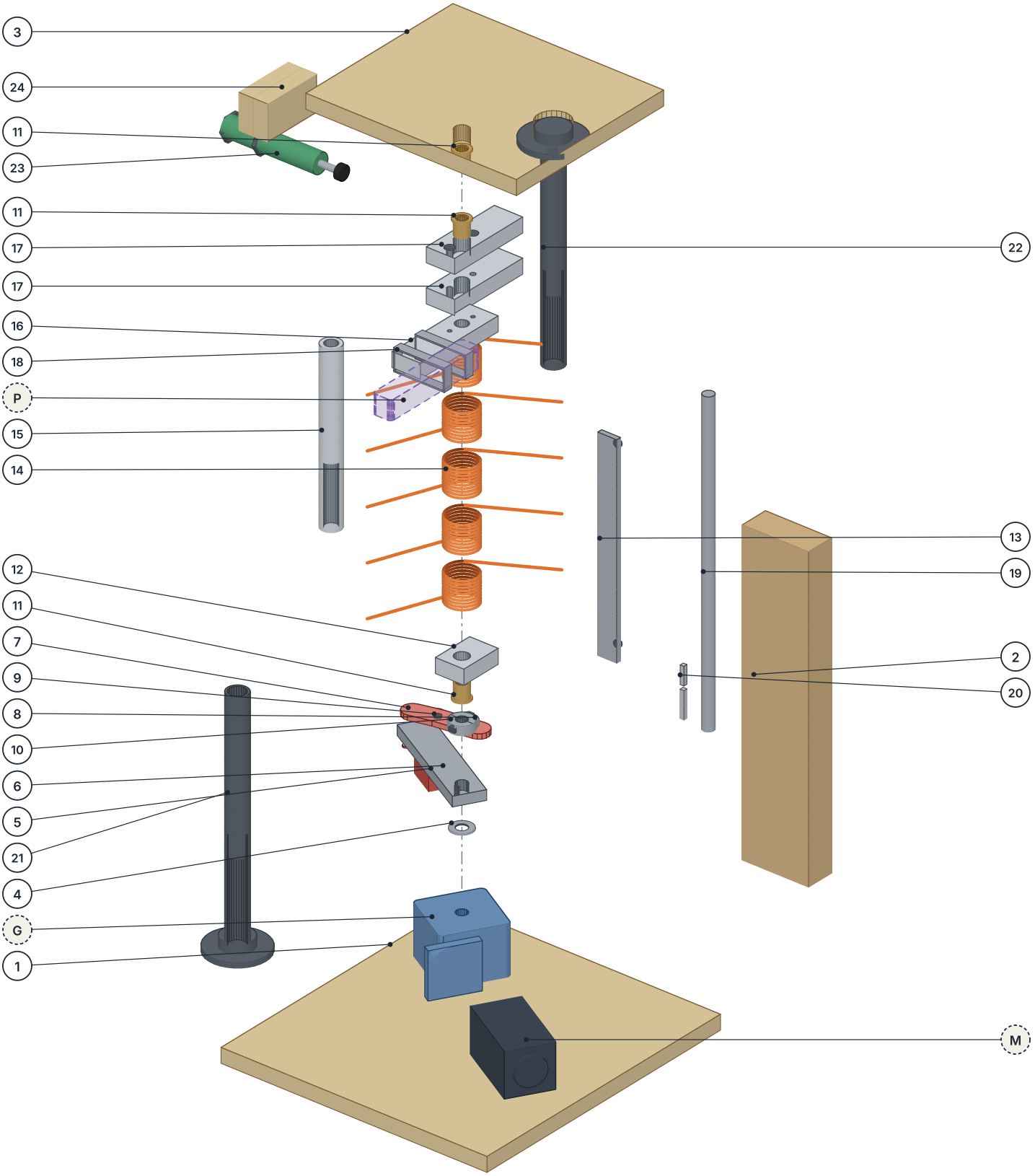
NOT IN THE COST

G	**Worm gearbox**	NMRV030 10:1 worm gearbox for NEMA 23, output up, motor lying flat
M	**NEMA 23 + driver**	About 2.5 N·m at the top setting; keep the driver enabled while cocked
P	**Paddle**	Handle clamped along the hub block, top face flush with the block top
	ESP32, power	24 V for the driver; 6 V 3 A buck for the servo

Directions in this document: **front** = where the paddle points at rest; clockwise / counterclockwise = seen from above. Cocking is clockwise.

Option 1 · torsion pack, exploded

DWG PL-01 · ISOMETRIC · NOT TO SCALE
5 springs on the shaft · 44 cm tall
BOM on the next page



KEY

Springs · Steel · Aluminum · Bronze bushings · Hook & servo · Absorber · Plywood / lumber · Gearbox & motor

Numbers = BOM items. Dashed letters = parts not in the bill of materials. Parts go on the shaft bottom to top in number order; the drive bar (13) and anchor pipe (21) go in last, see assembly.

PADDLE LAUNCHER · V1 FLAT SWING

Option 1 · bill of materials and where to buy

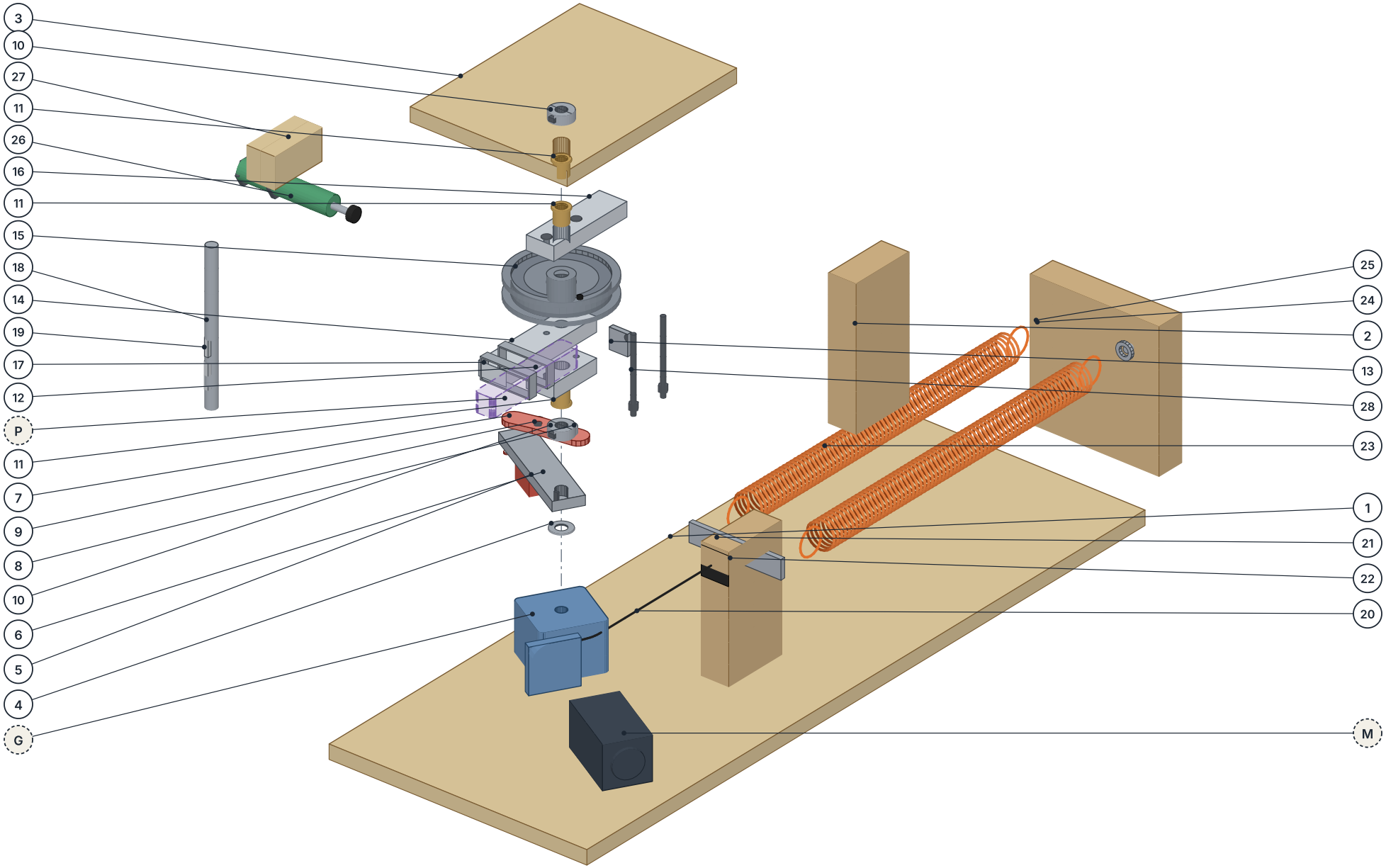
Links open the McMaster part or a store search
Everything listed ships today or is in stores today

#	PART	SIZE AND NOTES	QTY	WHERE TO BUY · HOW TO MAKE	EST.
FRAME					
1	**Base**	¾ in plywood, 14 × 15 in. 2¼ in hole for the gearbox spigot, 4 countersunk M6 holes matching the gearbox	1	Home Depot: ¾ in plywood 2 × 4 ft panel (also makes #3, #24) · page 8	$28
2	**Post**	2×4, 15¾ in (400 mm), back left, outside the paddle swing	1	Home Depot: 2×4 × 8 ft · page 8	$5
3	**Top plate**	¾ in plywood, 11⅛ × 9½ in. Ø18 mm bushing hole, 1⅝ in hole for the stop-pipe flange	1	From #1 plywood · page 8	—
RELEASE (SAME IN BOTH OPTIONS)					
4	**Washers**	M14 flat washers: one under the arm, 1–2 as shims under the top plate	3	In the hardware line	—
5	**Drive arm**	½ × 1½ in steel, 6¼ in, holes per sketch	1	McMaster: 1018 steel bar ½ × 1½ in, 1 ft · make it: page 6	$14
6	**Release servo**	MG996R metal-gear servo, hangs under the arm	1	Amazon: MG996R	$10
7	**Hook**	¼ × 1 in steel, 4⅛ in, 30° ramp on the nose	1	From the ¼ × 1 bar (#13) · page 6	—
8	**Hook roller**	608-2RS bearing on an M8 × 30 bolt with a nylock nut	1	Amazon: 608-2RS, 10-pack	$7
9	**Hook pivot + return spring**	M6 shoulder bolt (8 × 10 mm shoulder) + small extension spring	1+1	Amazon: shoulder bolt, spring kit	$12
PADDLE HUB AND SPRINGS					
10	**Shaft collar**	14 mm clamp collar under the rear block; sets the hub height	1	Amazon: 14 mm collar (2-pack)	$9
11	**Bronze bushings**	14 mm ID, 18 mm OD, 18 mm long, 22 mm flange. In #12, the top piece of #17, and #3	3	McMaster 6659K22	$11
12	**Rear block**	¾ × 1½ in 6061 aluminum, 2¼ in	1	McMaster: 6061 aluminum bar ¾ × 1½ in, 2 ft (makes #12, #16, #17) · page 6	$22
13	**Drive bar**	¼ × 1 in steel, 10½ in	1	Home Depot: Everbilt ¼ × 1 × 36 in steel flat bar (makes #7, #13) · page 6	$13
14	**Torsion springs**	270°, left-hand, 42.86 in·lbf, 1.666 in OD, 0.135 in wire, 4 in legs	5	McMaster 9271K132 (right-hand: 9271K131)	$48
15	**Spring mandrel**	1 in OD × ⅝ in ID aluminum tube, cut 8⅝ in	1	McMaster: 6061 tube, 1 ft · hacksaw, page 8	$11
16	**Front block (paddle hub)**	¾ × 1½ in aluminum, 5 in. Clearance hole, no bushing	1	From #12 bar · page 6	—
17	**Stop block**	2 pieces of ¾ × 1½ in aluminum, 4⅜ in each, stacked on #16. The absorber hits it	2	From #12 bar · page 6	—
18	**Hose clamps**	Stainless, 1½–2½ in: paddle handle to #16, handle top flush with the block top	2	Amazon: hose clamps	$5
19	**Keyed shaft**	14 mm × 400 mm, 5 mm keyway. As-is: 45 mm into the gearbox, top flush with #3	1	McMaster 1439K27	$26
20	**Keys**	5 × 5 mm key stock: 35 mm (gearbox) and 24 mm (arm)	2	McMaster: metric key stock. Skip if the gearbox came with a key	$6
SPRING STOPS AND ABSORBER					
21	**Anchor pipe**	¾ in black pipe nipple, 12 in, + floor flange on the base. The 5 bottom legs rest on it	1+1	Home Depot: nipple + flange	$15
22	**Stop pipe**	¾ in black nipple, 11 in, + floor flange under #3. The 5 top legs rest on it (holds the preload)	1+1	Zoro: ¾ × 11 in nipple (or McMaster); flange from Home Depot	$18
23	**Shock absorber**	AC2525-1: M25 × 1.5, 25 mm stroke, with 2 nuts. Hangs under #3, above the paddle	1	Amazon: AC2525-1 (the -1 is the high-speed version; see notes)	$16
24	**Absorber hanger**	2 layers of ¾ in plywood glued, 3⅛ × 1¾ in, 1 in hole. Glued and screwed under #3 + 2 corner braces	1	From #1 plywood · page 8	—
25	**Hardware (not drawn)**	4× M6 × 30 flat-head (gearbox), 4× M6 × 16 socket-head (drive bar), 2× M6 × 50 flat-head (#17), 6× 1 in ID nylon washers (between springs), 8 corner braces, #8 wood screws, wood glue, M6 tap + 5 mm drill	lot	Amazon or hardware store	$24
	Estimated total, everything new				**≈ $300**

≈ $243 with plywood, lumber and basic hardware on hand. Prices are estimates; check current prices before ordering. "page 6–8" = how to make that part from the stock you buy. The gearbox, motor, paddle, ESP32 and power supply are not included.

Option 2 · V-pulley drum and extension springs, exploded

DWG PL-02 · ISOMETRIC · NOT TO SCALE
2 springs behind the shaft · 24 cm tall
BOM on the next page



KEY Springs · Steel · Aluminum · Bronze bushings · Hook & servo · Absorber · Plywood / lumber · Gearbox & motor

Numbers = BOM items; letters = parts not in the bill of materials. Hub parts 12–16 stack on the shaft in that order; two long bolts (28) clamp them together.

Option 2 · bill of materials and where to buy

Links open the part or a store search
Century Spring ships stock springs from their site

#	PART	SIZE AND NOTES	QTY	WHERE TO BUY · HOW TO MAKE	EST.
FRAME					
1	**Base**	¾ in plywood, 14 × 37 in. 2¼ in hole for the gearbox spigot + 4 countersunk M6 holes	1	Home Depot: ¾ in plywood 2 × 4 ft panel (also makes #3, #27) · page 8	$28
2	**Post**	2×4, 7¼ in (185 mm), back right	1	Home Depot: 2×4 × 8 ft (also makes #22) · page 8	$5
3	**Top plate**	¾ in plywood, 8½ × 11¼ in, Ø18 mm bushing hole	1	From #1 plywood · page 8	—
RELEASE (SAME IN BOTH OPTIONS)					
4	**Washers**	M14 flat washers: one under the arm, 1–2 as shims under the top plate	3	In the hardware line	—
5	**Drive arm**	½ × 1½ in steel, 6¼ in, holes per sketch	1	McMaster: 1018 steel bar ½ × 1½ in, 1 ft · make it: page 6	$14
6	**Release servo**	MG996R metal-gear servo, hangs under the arm	1	Amazon: MG996R	$10
7	**Hook**	¼ × 1 in steel, 4⅛ in, 30° ramp on the nose	1	From the ¼ × 1 bar (#13) · page 7	—
8	**Hook roller**	608-2RS bearing on an M8 × 30 bolt with a nylock nut	1	Amazon: 608-2RS, 10-pack	$7
9	**Hook pivot + return spring**	M6 shoulder bolt (8 × 10 mm shoulder) + small extension spring	1+1	Amazon: shoulder bolt, spring kit	$12
PADDLE HUB AND DRUM (ON THE SHAFT, BOTTOM TO TOP)					
10	**Shaft collars**	14 mm: one under the rear block, one on top of #3	2	Amazon: 14 mm collar (2-pack)	$9
11	**Bronze bushings**	14 mm ID, 18 mm OD, 18 mm long, 22 mm flange. In #12, #16 and #3	3	McMaster 6659K22	$11
12	**Rear block**	¾ × 1½ in 6061 aluminum, 2⅞ in	1	McMaster: 6061 aluminum bar ¾ × 1½ in, 2 ft (makes #12, #14, #16) · page 7	$22
13	**Drive lug**	¼ × 1 in steel, 1⅛ in; the hook roller pushes on it	1	Home Depot: Everbilt ¼ × 1 × 36 in steel flat bar (makes #7, #13, #21) · page 7	$13
14	**Front block (paddle hub)**	¾ × 1½ in aluminum, 6 in. Clearance hole, no bushing	1	From #12 bar · page 7	—
15	**V-pulley (drum)**	5 in OD single-groove A pulley, ⅝ in bore, die-cast (Chicago Die Cast 500A6). Drill 2 bolt holes + 1 cord hole	1	Zoro 500A6; also at Ace / True Value · page 8	$18
16	**Stop block**	¾ × 1½ in aluminum, 4⅞ in, on top of the pulley. The absorber hits it	1	From #12 bar · page 7	—
17	**Hose clamps**	Stainless, 1½–2½ in: paddle handle to #14, handle top flush with the block top	2	Amazon: hose clamps	$5
18	**Keyed shaft**	14 mm × 200 mm, 5 mm keyway, as-is	1	McMaster 1439K26	$20
19	**Keys**	5 × 5 mm key stock: 35 mm and 24 mm	2	McMaster: metric key stock	$6
SPRING TRAIN (BEHIND THE SHAFT, WHERE THE PADDLE NEVER GOES)					
20	**Cord**	3 mm Dyneema (UHMWPE) cord, about 1 m used. Pulley to equalizer	1	Amazon: 3 mm Dyneema	$12
21	**Equalizer**	¼ × 1 in steel, 5 in, 3 holes	1	From #13 bar · page 7	—
22	**Spring stop**	2×4 on end, 6½ in, between the springs; ⅛ in rubber pad on its front face (holds the preload)	1	From #2 2×4 + rubber sheet	$5
23	**Extension springs**	Century Spring 660CS: 1.5 in OD, 0.135 in wire, 11.5 in free, 2.7 lb/in, 47 lb max	2	centuryspring.com 660CS (stock part, ships from Century)	$25
24	**Quick links + eye bolts**	M6 quick links (4 at the springs, 1 for the cord); 5/16 × 4 in eye bolts with nuts and washers	5+2	Amazon: quick links; Home Depot: eye bolts	$14
25	**Anchor block**	2×8, 7 in, on edge across the base, 2 corner braces behind	1	Home Depot: 2×8 (or any 7 in offcut) · page 8	$10
ABSORBER AND HARDWARE					
26	**Shock absorber**	AC2525-1: M25 × 1.5, 25 mm stroke, with 2 nuts. Hangs under #3, above the paddle	1	Amazon: AC2525-1 (the -1 is the high-speed version; see notes)	$16
27	**Absorber hanger**	2 layers of ¾ in plywood glued, 3⅛ × 1⅝ in, 1 in hole. Glued and screwed under #3 + 2 corner braces	1	From #1 plywood · page 8	—
28	**Hardware**	2× M6 × 100 flat-head + nylocks (through the hub; drawn loose), 2× M6 × 16 socket-head (lug), 4× M6 × 30 flat-head (gearbox), 8 corner braces, #8 wood screws, glue, M6 tap + 5 mm drill	lot	Amazon or hardware store	$24
	Estimated total, everything new				**≈ $286**

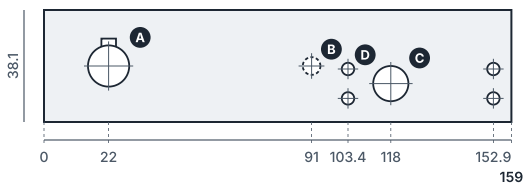
≈ $219 with plywood, lumber and basic hardware on hand. Prices are estimates; check current prices before ordering. "page 6–8" = how to make that part from the stock you buy. The gearbox, motor, paddle, ESP32 and power supply are not included.

Making the cut-from-stock parts · release + option 1

Arm and hook are shared by both options
Option 2's parts are on the next page

#5 Drive arm

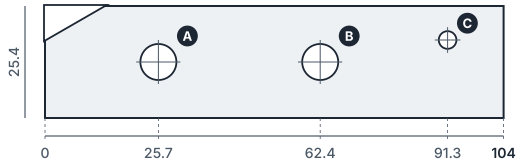
Buy: McMaster 1018 steel bar ½ × 1½ in, 1 ft · **Cut:** cut 6¼ in (159 mm). Dimensions in mm from the 0 end.



A Ø14 mm hole with a 5 mm wide × 2.3 mm deep notch: drill Ø14, then file the notch until the key slides in snugly
B M6 thread for the hook pivot bolt: drill 5 mm, tap M6
C Ø12 mm clearance hole for the servo output shaft (servo hangs underneath)
D 4 × Ø4.2 mm for the servo ears; check against the servo before drilling

#7 Hook

Buy: the Home Depot ¼ × 1 in flat bar · **Cut:** cut 4⅛ in (104 mm). Dimensions in mm from the 0 end.



A Ø8.2 mm: M8 bolt + 608 bearing (the roller) on the side facing up
B Ø8.2 mm: pivot, for the 8 mm shoulder of the M6 shoulder bolt
C Ø4 mm: return-spring hole
Grind a 30° ramp on the nose corner (top left as drawn) so the drive bar can push the hook open on reset

#13 Drive bar (option 1)

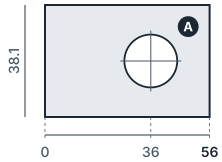
Buy: the same ¼ × 1 in flat bar · **Cut:** cut 10½ in (267 mm). Dimensions in mm from the 0 end.



A 4 × Ø6.6 mm, pairs 15 mm apart across the width: M6 × 16 screws into the rear and front blocks
Hook end is the 0 end. File the long edges smooth: the spring legs and the hook roller run on them

#12 Rear block (option 1)

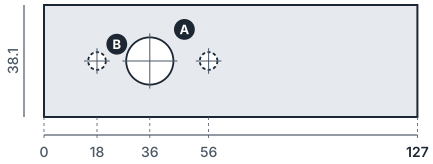
Buy: McMaster ¾ × 1½ in 6061 bar, 2 ft · **Cut:** cut 2¼ in (56 mm). Dimensions in mm from the 0 end.



A Ø18 mm bore for a bushing (see the shop tips on page 7)
In the 0 end face: 2 × M6 threads, 15 mm apart, centered (drive bar screws)

#16 Front block / paddle hub (option 1)

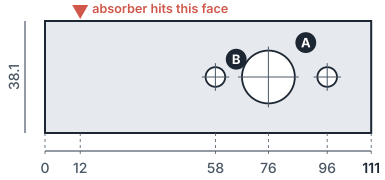
Buy: the same aluminum bar · **Cut:** cut 5 in (127 mm). Dimensions in mm from the 0 end.



A Ø16 mm clearance hole (no bushing here)
B 2 × M6 threads straight through the block (drill 5 mm, tap M6): the stop block screws into these
In the 0 end face: 2 × M6 threads, 15 mm apart (drive bar screws). The paddle handle clamps along the long side, flush with the top

#17 Stop block, 2 pieces (option 1)

Buy: the same aluminum bar · **Cut:** cut 2 pieces, 4⅜ in (111 mm) each. Dimensions in mm from the 0 end.



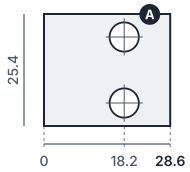
A Top piece: Ø18 mm bore for a bushing, flange up. Bottom piece: Ø16 mm clearance instead
B 2 × Ø6.6 mm through both pieces; countersink them on the top piece for M6 × 50 flat-head screws
Stack the two pieces on the front block, 0 ends toward the back

Making the cut-from-stock parts · option 2 + shop tips

Arm (#5) and hook (#7) are on the previous page

#13 Drive lug (option 2)

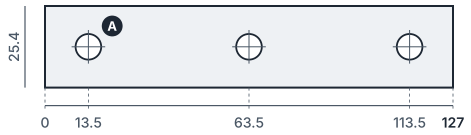
Buy: the ¼ × 1 in flat bar · **Cut:** cut 1⅛ in (28.6 mm). Dimensions in mm from the 0 end.



A 2 × Ø6.6 mm, 15 mm apart: M6 × 16 into the rear block end. The hook roller pushes on its edge

#21 Equalizer (option 2)

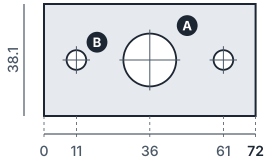
Buy: the ¼ × 1 in flat bar · **Cut:** cut 5 in (127 mm). Dimensions in mm from the 0 end.



A 3 × Ø8 mm, 50 mm apart: springs on the outer holes (via quick links), cord tied to the middle one

#12 Rear block (option 2)

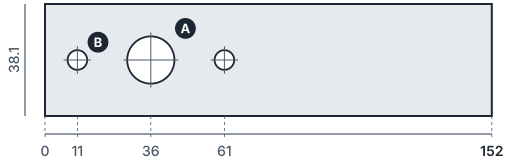
Buy: McMaster ¾ × 1½ in 6061 bar, 2 ft · **Cut:** cut 2⅞ in (72 mm). Dimensions in mm from the 0 end.



A Ø18 mm bore for a bushing
B 2 × Ø6.6 mm for the M6 pulley bolts
In the 0 end face: 2 × M6 threads, 15 mm apart (drive lug screws)

#14 Front block / paddle hub (option 2)

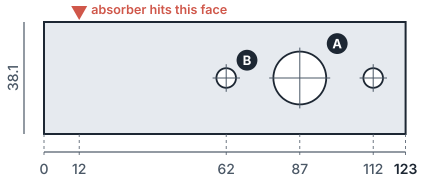
Buy: the same aluminum bar · **Cut:** cut 6 in (152 mm). Dimensions in mm from the 0 end.



A Ø16 mm clearance hole (no bushing here)
B 2 × Ø6.6 mm for the M6 pulley bolts
The paddle handle clamps along the long side, flush with the top

#16 Stop block (option 2)

Buy: the same aluminum bar · **Cut:** cut 4⅞ in (123 mm). Dimensions in mm from the 0 end.



A Ø18 mm bore for a bushing, flange up
B 2 × Ø6.6 mm, countersunk on top for the M6 × 100 flat-head pulley bolts
Sits on top of the pulley, 0 end toward the back

Shop tips for every metal part

Mark and cut: mark with a square and scribe, centre-punch every hole. Cut with a hacksaw (or a bandsaw) and file the ends square.

Drill: a drill press makes this much easier. Clamp the part, pilot 3 mm first, then the final size. Cutting oil on steel, slow speed on the ½ in arm. Deburr both sides.

Tap M6: drill 5 mm, tap with oil, back the tap off every half turn. Aluminum taps easily.

Bushing bores (Ø18): drill 17.5 mm (or 11/16 in), then 18 mm. Press the bushing in with a vise and a socket. If it slides in loose, glue it with Loctite 638 or epoxy. Check the shaft turns freely afterwards.

Key notch on the arm: drill Ø14, then file the notch using the key itself as a gauge until it slides in snugly.

Countersinks: a 90° countersink bit in the drill press, until an M6 flat-head screw sits flush.

Wood parts, store parts you modify, and tools

All hole positions are from the shaft centre or the part edges
Front = paddle direction at rest

Wood parts · option 1

1	**Base**	¾ ply, 14 × 15 in	Shaft centre 7 in from the left edge and 176 mm from the front edge: 2¼ in hole + gearbox screws. Anchor flange 62 mm right, 66 mm in front of the shaft. ½ in hole for the temporary rod 35 mm right, 84 mm in front.
2	**Post**	2×4, 15¾ in	Wide face toward the shaft, centre 60 mm left and 171 mm behind the shaft. 2 corner braces at each end.
3	**Top plate**	¾ ply, 11⅛ × 9½ in	Ø18 mm bushing hole 110 mm from the left edge, 45 mm from the front edge. 1⅝ in hole (stop-pipe flange boss) 62 mm left and 74 mm behind the bushing hole.
24	**Absorber hanger**	2 × ¾ ply, 3⅛ × 1¾ in	Glue face to face first, then drill a 1 in hole centred, 21 mm up from the bottom edge. Sits under the plate's right edge.

Wood parts · option 2

1	**Base**	¾ ply, 14 × 37 in	Shaft centre 7 in from the left edge and 174 mm from the front edge: 2¼ in hole + gearbox screws.
2	**Post**	2×4, 7¼ in	Long side front-to-back, centre 60 mm right and 190 mm behind the shaft.
3	**Top plate**	¾ ply, 8½ × 11¼ in	Ø18 mm bushing hole 45 mm from the left edge and 45 mm from the front edge.
22	**Spring stop**	2×4, 6½ in	Stands on end, centre 55 mm left of the shaft line; front face 191 mm behind the shaft. Rubber pad on the front face.
25	**Anchor block**	2×8, 7 in	On edge across the base, front face 710 mm behind the shaft. Two 5/16 in holes 100 mm apart, centred 55 mm left of the shaft line, 145 mm up.
27	**Absorber hanger**	2 × ¾ ply, 3⅛ × 1⅝ in	As option 1: 1 in hole 21 mm up from the bottom edge.

The exact spot for each hanger is wherever the absorber cap just touches the stop block with the hub at rest. Mark it with the plate in place.

Store-bought parts you modify

1. **V-pulley (option 2, #15).** Drill two 6.6 mm holes through the web, 25 mm either side of the centre, on one line. With that line pointing front-to-back, drill a 5 mm cord hole straight down through the top rim at the bottom of the groove, 60° round from the front hole toward the left. The drawings assume the pulley is about 1⅛ in (28 mm) thick. If the pulley is thicker, everything above it moves up by the difference.
2. **Mandrel (option 1, #15).** Cut the 1 in tube to 8⅝ in with a hacksaw, deburr the bore.
3. **Keys.** Cut the 5 × 5 mm key stock to 35 mm and 24 mm.
4. **Cord (option 2, #20).** Dyneema is slippery. At the pulley, a figure-8 stopper knot on top of the rim with a 50 mm tail. At the equalizer, pass it twice through the quick link and tie a bowline backed with a double overhand. A drop of thin CA glue on each knot.
5. **Pipes, shaft, springs, absorber, clamps:** used as bought.

Tools

Hacksaw or bandsaw · drill (a drill press strongly recommended) · bits 3, 4.2, 5, 6.6, 8, 8.2, 12, 14, 16, 17.5, 18 mm (a step drill covers several) · 1 in spade or Forstner bit · 2¼ in and 1⅝ in hole saws · 90° countersink · M6 tap and handle · files · centre punch · square · clamps and a vise · circular saw or jigsaw for the plywood.

Rather not cut metal yourself?

Plywood: Home Depot and Lowe's will make straight cuts on a panel in the store, sometimes for a small fee. Bring the sizes above; drill the holes at home.

Bars cut to length: OnlineMetals and Metals Depot cut bar to length, and McMaster offers cut-to-length on some bars. You still drill the holes.

Finished flat parts: SendCutSend or OSH Cut laser- or waterjet-cut parts from DXF files, holes included. Countersinking and tapping are add-on services, and parts typically arrive in about a week. Good fits are the arm, hook, drive bar, lug and equalizer.

Local shop or makerspace: pages 6–8 are enough for a machine shop to quote from.

Specialized parts and where they come from

Torsion springs (option 1): no single stock torsion spring is strong enough. The biggest stock ones top out around 4.8 N·m, so five identical McMaster springs share the load. One custom spring would replace them but takes weeks to make.

Extension springs (option 2): 660CS is a stock Century Spring part sold from their website. If it's out of stock, any extension spring around 1½ in OD and 11–12 in long works if it is about 2.7 lb/in and stretches to 23.8 in (604 mm) carrying at least 42 lb.

Assembly order

Order matters for the springs in both options

Option 1 · torsion pack

1. Make the parts (pages 6–8). Press bushings into the rear block, the top stop piece (flange up) and the top plate (flange up).
2. Bolt the gearbox under the base (spigot in the 2¼ in hole, flat-head M6 from below); the motor lies flat, pointing front-left. Screw down the anchor-pipe **flange only**, and the 2×4 post at the back left.
3. Long key in the gearbox bore, shaft down into it (45 mm). Add a washer, the arm (key, servo and hook fitted) and the collar.
4. Slide on the rear block, then the mandrel and the five springs with a nylon washer between each. Leave the springs loose.
5. Screw the two stop pieces onto the front block (M6 × 50 flat-heads into its tapped holes) and slide that unit onto the shaft.
6. Screw the stop-pipe flange under the top plate (boss up in the 1⅝ in hole) and thread in the 11 in nipple. Lower the plate over the shaft, add M14 washers until the hub has about 1 mm of up-down play, and brace the plate to the post.
7. Turn each spring on the mandrel until its **top** leg rests against the stop pipe, pushing counterclockwise. No force needed yet.
8. Stand a temporary ½ in rod in its hole. One spring at a time, slip a short pipe over the **bottom** leg and swing it counterclockwise about 145° (about 6 lb at the leg end) until it rests behind the rod. Then lower the 12 in anchor nipple in from above and screw it into its flange. Pull the rod out: the legs settle on the anchor pipe.
9. Slide the drive bar in from the back, into the gap just counterclockwise of the springs' top legs, and screw it to the rear and front blocks (M6 × 16).
10. Glue and screw the hanger under the plate with the absorber in it. Set the cap to just touch the stop block with the bar about 10° short of the springs' top legs.
11. Clamp the paddle (top flush with the hub block). Cock slowly with the motor and check every clearance, then the first shot at the lowest setting.

Why the anchor pipe goes in last: each spring has to be wound past one of its two stops, and the anchor pipe is the one you can drop in from above afterwards. The temporary rod hole is in the wood-parts table on page 8.

Option 2 · extension springs

1. Make the parts (pages 6–8). Press bushings into the rear block, the stop block (flange up) and the top plate (flange up). Drill the pulley.
2. Bolt the gearbox under the base. Screw down the post (back right), the spring stop and the anchor block with corner braces; fit the eye bolts.
3. Long key in the gearbox bore, shaft down into it. Add a washer, the arm (key, servo and hook fitted) and a collar.
4. Stack on the shaft: rear block (lug screwed to its end), front block, pulley, stop block. Drop the two M6 × 100 flat-heads down through all four and tighten the nylocks underneath.
5. Thread the cord up through the pulley's cord hole and tie the stopper knot on top. Lead it counterclockwise round the groove to the pulley's left side, then straight back.
6. Lower the top plate over the shaft, shim to about 1 mm of hub play, screw it to the post, and fit the top collar.
7. Quick-link each spring's back loop to its eye bolt and put quick links in the equalizer's outer holes. Hook a ratchet strap from the equalizer to the front of the base and stretch the springs about 8 in (30 lb each). Close the quick links into the springs' front loops, then let the strap off slowly; the equalizer settles on the spring stop's pad.
8. With the paddle hub at rest, pull the cord just snug, let out 10 mm (this is the 10° slack), and tie it to the middle quick link.
9. Glue and screw the hanger under the plate with the absorber in it; set the cap to just touch the stop block with the hub at rest.
10. Clamp the paddle (top flush with the hub block). Cock slowly with the motor and check every clearance, then the first shot at the lowest setting.

At rest the two springs pull about 60 lb on the spring stop. Keep the ratchet strap on whenever you work near the spring train.

Numbers, firmware settings and notes

Steps assume 1600 steps per motor turn and 10:1 (44.4 steps per degree)
Slack 10° before the springs pick up

Option 1 · numbers

Spring pack	5.14 N·m/rad (0.090 N·m per degree), 5 × 9271K132
Preload at rest	11.3 N·m: each spring wound 126° onto the stop pipe
Top setting	21.2 N·m at 236°, 87% of the springs' 270° rating
Hub inertia	paddle 0.022 + hub 0.0022 kg·m²
Hook load	about 540 N at the top setting

Option 2 · numbers

Springs	2 × 660CS on a 55 mm cord radius: 2.86 N·m/rad
Preload at rest	14.7 N·m: each spring stretched to 498 mm (134 N, 30 lb)
Top setting	20.2 N·m; each spring at 604 mm and 184 N (41 lb), 87% of its 47 lb maximum
Hub inertia	paddle 0.022 + hub 0.0029 kg·m²
Hook load	about 516 N at the top setting

Option 1 · firmware settings

AVG ACCEL °/S²	TIP M/S	COCK ANGLE	MOTOR STEPS	SPRING N·M	MOTOR N·M
7,000	3.1	13.8°	612	11.6	1.37
10,000	4.0	16.3°	725	11.9	1.40
15,000	5.9	23.0°	1,024	12.5	1.47
20,000	8.5	35.7°	1,588	13.6	1.60
25,000	12.2	59.0°	2,621	15.7	1.85
30,000	16.8	93.7°	4,166	18.8	2.21
33,200	20.0	120.0°	5,333	21.2	2.49

Option 2 · firmware settings

AVG ACCEL °/S²	TIP M/S	COCK ANGLE	MOTOR STEPS	SPRING N·M	MOTOR N·M
7,000	3.0	13.0°	576	14.9	1.75
10,000	3.9	14.8°	658	15.0	1.76
15,000	5.4	19.4°	861	15.2	1.79
20,000	7.4	27.5°	1,223	15.6	1.83
25,000	10.5	44.3°	1,968	16.4	1.93
30,000	15.7	81.3°	3,614	18.3	2.15
33,200	20.0	120.0°	5,333	20.2	2.38

Check before you order

1. **Gearbox output:** NMRV030 units normally have a 14 mm hollow bore with a 5 mm keyway, so the McMaster shaft drops straight in. A gearbox with a solid output shaft needs a different shaft line.
2. **Motor flange:** one buyer of a similar NMRV030 found its NEMA 23 holes on 49 mm centres instead of 47.1 mm. Check that the motor bolts up before cutting the base.
3. **Absorber speed:** the stop block reaches the absorber at about 3.3 m/s (option 1) and 3.8 m/s (option 2). Cheap AC2525s come in -1 / -2 / -3 versions for high / medium / low speed; one -3 listing is rated to only 1.0 m/s. Buy the -1. The sturdier upgrade is an SMC RBC2725 (M27, rated to 5 m/s; drill the hanger 1⅛ in). Either way the stop swings about 20° while it pushes, so the rod takes some side load. Expect to replace a cheap absorber now and then.
4. **Cock direction:** the torsion springs are left-hand for a paddle that cocks clockwise seen from above. For the other way, order 9271K131 and mirror the layout.
5. **Paddle height:** the absorber hangs 3 mm above the paddle's top face, so clamp the handle flush with or below the hub block's top.

Safety

Both options store about 31 J at the top setting, and both are preloaded even at rest. Keep hands out of the swing, fit a guard or a clear zone, and never dry-fire above the lowest setting until everything is checked.

Cost

About $300 (option 1) and $286 (option 2) new, or $243 and $219 with plywood, lumber and hardware on hand. The biggest items are the springs, the metal stock (~$50) and the shaft.